

TEAM TIBURON SOFTWARE INDUCTION

Welcome to team tiburon software inductions. You can **choose any 2 of the below given tasks to do and submit before 10/4/2026 6:00PM.**

TASKS

Tasks will be **40%** of the marks.

1. Turtle Simulation

- **Shape Drawing:** draw square, triangle, cube shape using turtlesim
- **Navigation:** Navigate through different positions that can be given at runtime.
- **GUI Services:** Use Python **Tkinter** or **PyQt** to create buttons that call ROS2 services like **reset**, **clear**, or **spawn** turtles or change color of pen of turtle
- **Follow the Girl:** Spawn a turtle. Name a turtle girl. make it move. And make another turtle follow the first one.

2. Gazebo Ignition Fortress

- **Buoyancy Slider:** Find a plugin to create buoyancy. And control the buoyancy using a slider GUI.
- **Multi-Bot Spawning:** Spawn multiple robots in Gazebo. Find or make the URDF.

3. ROS2 Communication

Managing data flow and timing.

- **Frequency Control:** to ensure the publisher maintains a Frequency.
- **Data Sync:** align messages from multiple sensors (e.g., IMU and Camera) based on their header timestamps.

4. Perception

- **Edge Detection:** Implement the edge detector operator to locate edges in specific directions in **x** or **y**.

- **Color Tracking:** Track a particular color from the camera stream and provide the centroid of the color to the terminal. Color as per your preference.

EVALUATION

- Create a github repo and submit the project as **Public**.
- Make a detailed Report of the entire Projects and upload to github.
- The report should contain **Personal Details** (Name, Roll No, Contact information, Linkedin, Github, email), **Project Report**.
- Project report should contain detailed analysis of **what the problem is? How did you plan to solve it? What are the ways to solve the problem? What are the tradeoffs to different approaches to solving a problem? If any.**
- **Project report max 5 pages.**
- Make a ReadMe for the github page. The readme should contain a summary and a user guide. And video or gif of the solution. **Make the readme very short**